

EDUCATION

University of Toronto

PhD Candidate, Electrical and Computer Engineering (ECE)

*Toronto, Canada
Sep 2021 — Present*

- **Thesis:** Near Field Wireless Power Transfer and Focusing using Electromagnetic Metamaterials
- Using numeric methods, optimization, and finite-element simulation tools to design and optimize relevant antenna structures, e.g. COMSOL, HFSS, and the method of moments (MoM). Designs are fabricated in-lab using laser printed circuit board (PCB) prototyping machine. Experimental verification is performed using relevant RF measurement setups.
- **Available for full-time positions starting Nov 2026.**

University of Toronto

Master of Applied Science (MAsc), Electrical and Computer Engineering (ECE)

*Toronto, Canada
Sep 2019 — Sep 2021*

- **Thesis:** Shifted-beam Array of Coils for Highly Focal Transcranial Magnetic Stimulation, [link](#)

University of Toronto

Honors Bachelors of Science (BSc Hons), Physics & Mathematics Double Major

*Toronto, Canada
Sep 2015 — Jun 2019*

- Dean's List Award recipient 2016-2018.

TECHNICAL SKILLS

Programming Languages:	Python, MATLAB, C++, Java
Simulation Tools:	COMSOL Multiphysics, Ansys Electronics Desktop (AEDT: HFSS, Icepak), Keysight Advanced Design System (ADS)
Modeling/Numerical Methods:	Finite-element method (FEM), finite-difference time-domain (FDTD), method of moments (MoM), reduced order modeling/surrogate modeling, multiphysics modeling, optimization
AI/ML Tools:	Tensorflow, Keras, PyTorch
Software Tools:	Anaconda, Jupyter Notebook, PyCharm, VSCode, LaTeX, Git, high performance computing (HPC)
Subject Areas:	Electromagnetics, metamaterials, physics, wireless power transfer
Hardware:	Radiofrequency (RF) design, RF equipment, antenna design/measurements, printed circuit board (PCB) prototyping, laser etching, 3-D printing, composite materials, machining

PROFESSIONAL EXPERIENCE

Research & Development Intern

Synopsys: Electronics, Simulation, and Optics Unit

*Remote, Canada
May 2025 - Apr 2026*

- **Developed, implemented, and tested** methods for enhancing the prediction accuracy of surrogate thermal models for integrated circuit (IC) packages, **using a physics-informed data sampling approach**. Performed detailed study of effects with respect to dataset size, data quality, convergence, etc. **Achieved 60-80% error reduction** for tested IC packages compared to conventional surrogate modeling techniques.
- **Developed general-purpose data sampling workflow** for surrogate model training based on observed package sensor data, allowing accurate inference of underlying physics behavior. **Achieved 60-80% error reduction** for tested IC package.
- **Developed Bayesian inference analysis pipeline** to create robust, adaptive data sampling method capable of responding in real-time to live IC sensor data. Invention disclosure/patent in progress.
- Leveraged **PyCharm for code development**.

Researcher, Computational Physics/Atomistic Simulation

Toronto SickKids Hospital: SickKids Research Institute

*Toronto, Canada
Jan 2015 — Sep 2019*

- **Numerically simulated atomic interactions and protein chemical structures using molecular dynamics** on the SciNet supercomputer cluster. **Analyzed, preprocessed, and visualized large-scale datasets using Python**. Studied small brain peptides and interactions with amyloid-beta oligomers in the context of Alzheimer's disease.
- **Identified and categorized distinct preferential binding modes** consistent with previous literature on amyloid-beta oligomer behavior. Characterized binding propensities using statistical methods.

Research Assistant, Computational Solid State Physics

University of Toronto: Department of Physics

*Toronto, Canada
May 2018— Aug 2018*

- **Constructed new simplified tight-binding model of nickel (II) iodide** by calculated tight-binding parameters for electron-hopping model. Performed numeric simulation and data analysis of electron-hopping models using Python via Monte Carlo simulation and statistical methods.
- **Generated datasets from density functional theory (DFT) using HPC tools** on SciNet supercomputer cluster.

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- **Used Python to calculate energy band structure from new tight-binding model** and confirm agreement with experimental data from literature. Identified distinct magnetic ordering cases, including helimagnetic ordering, via simulated annealing using Monte-Carlo methods.

Research Assistant, Data Quality and Output

Sudbury, Canada

SNOLAB: SNO+ Experiment

May 2017— Aug 2017

- **Developed and debugged C++ scripts for data quality checks** on SNO+ detector output ROOT and relevant physics libraries.
- **Assembled and tested calibration apparatus** for SNO+ detector per clean room protocols. Assisted with equipment assembly in ultra-clean underground facility at 2 km depth.

Research Assistant, Atomic Physics

Toronto, Canada

University of Toronto: Department of Physics

May 2016— Jun 2016

- **Used RF equipment to test a small solid-state NMR apparatus** to measure precession frequencies of magnetic dipole moments in Pb-207 nuclei, in the context of searching for violations in time reversal symmetry.
- **Designed and built RF filters** and relevant RF setup for experiment.

ACADEMIC PUBLICATIONS

Wireless Power Transfer Using a Cavity-Enclosed Impedance Metawire Structure

2026 IEEE International Symposium on Antennas and Propagation, Detroit, United States

An Aperiodic Sub-Wavelength Dipole Structure for Enhancing Near-Field Wireless Power Transfer, [link](#)

IEEE Antennas and Wireless Propagation Letters (AWPL), Vol. 23, No. 6, July 2024

A Shifted-Beam Method for Near-Field Wireless Power Transfer using Parasitic Arrays, [link](#)

2023 IEEE International Symposium on Antennas and Propagation, Portland, United States

Shifted-Beam Array Coil for Highly Focal Transcranial Magnetic Stimulation, [link](#)

2021 IEEE International Symposium on Antennas and Propagation, Singapore

ORGANISATIONS AND VOLUNTEERING

Vice Chair, University of Toronto IEEE Antenna and Propagation Society (AP-S) Student Chapter

Oct 2023 — Dec 2025

- Hosted social and academic events for U of Toronto IEEE AP-S student chapter. Elected position.

Team Lead & Composites Fabricator, Rocketry Division, University of Toronto Aerospace Team (UTAT)

Sep 2015 — Aug 2021

- Led a team of more than 10 members for rocketry airframe subsystem and solid-fuel rocket program (2019-2021).
- Used machine tools and composite manufacturing techniques to design and fabricate by hand carbon fiber airframes for low-altitude sounding rockets. Performed compression testing to verify performance. Coordinated design and construction of solid-fuel rockets for systems testing.

CITIZENSHIP/LEGAL STATUS

- Canadian citizen.